

FRIDAY HARBOR, WA, USA

WMO: 727985

Lat: 48.522N

Lon: 123.023W

Elev: 33

StdP: 100.93

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 94276

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.1	-1.5	-12.4	1.3	-0.6	-9.0	1.8	0.6	13.7	7.8	12.4	7.6	2.1	10	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	26.4	17.1	24.0	16.5	22.3	15.8	17.8	25.1	16.8	23.2	16.1	21.7	2.9	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.7	10.5	19.2	14.0	10.0	18.6	13.4	9.6	18.0	49.7	25.1	47.1	23.3	44.9	21.6	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	8.4	6.9	-6.8	31.0	2.6	2.3	-8.7	32.6	-10.2	33.9	-11.7	35.2	-13.6	36.9		
(5)			-7.8	19.6	2.4	0.9	-9.6	20.3	-11.0	20.8	-12.4	21.2	-14.2	21.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	5.3	5.3	7.0	9.0	11.7	14.1	15.8	16.0	14.0	10.3	7.1	5.0
	DBStd	4.68	2.89	2.73	2.47	2.25	2.29	2.17	2.10	1.92	2.05	2.19	2.89	2.90
	HDD10.0	697	146	131	96	44	9	1	0	0	0	22	92	157
	HDD18.3	3026	403	364	351	279	206	131	85	77	129	248	336	414
	CDD10.0	727	1	1	3	14	61	122	179	186	122	32	6	1
	CDD18.3	14	0	0	0	0	0	3	6	5	1	0	0	0
	CDH23.3	276	0	0	0	1	6	48	107	94	21	0	0	0
	CDH26.7	55	0	0	0	0	1	10	26	17	2	0	0	0
Wind	WSAvg	2.4	2.9	2.8	2.8	2.5	2.1	2.2	2.0	1.7	1.6	2.0	2.8	3.1
Precipitation	PrecAvg	846	128	80	77	48	36	30	18	28	40	94	135	132
	PrecMax	1068	214	187	146	97	74	63	41	91	119	272	283	209
	PrecMin	628	59	13	23	5	2	6	1	4	3	14	48	51
	PrecStd	120	40	35	27	22	19	14	11	22	30	53	62	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	12.4	16.3	20.1	23.8	28.2	29.5	28.7	26.1	18.8	15.1	12.3
		MCWB	9.8	8.4	10.7	12.6	15.2	17.0	18.5	17.9	16.4	12.7	11.7	10.1
	2%	DB	11.1	11.2	13.5	17.1	20.9	24.1	26.2	26.0	22.8	16.9	13.1	10.9
		MCWB	9.2	8.1	9.3	11.3	14.1	16.1	17.4	17.2	15.3	12.2	11.1	9.1
	5%	DB	10.1	10.1	12.2	14.9	18.5	21.4	23.7	23.6	20.9	15.3	12.2	9.7
		MCWB	8.4	7.6	8.6	10.0	13.2	15.3	16.5	16.6	14.8	11.8	10.4	8.0
	10%	DB	9.1	9.1	11.0	13.3	16.9	19.0	21.7	21.8	18.9	14.0	11.2	8.8
		MCWB	7.5	7.1	8.1	9.3	12.4	14.1	15.8	15.9	14.2	11.3	9.5	7.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	10.3	11.5	13.5	16.0	17.9	19.3	18.6	17.4	14.8	12.7	11.0
		MCDB	11.5	11.3	14.5	18.7	22.0	25.7	28.0	26.5	23.4	16.7	13.7	11.7
	2%	WB	9.7	9.0	10.1	11.8	14.7	16.5	17.8	17.7	16.2	13.3	11.7	9.3
		MCDB	10.7	10.2	12.4	16.0	19.9	23.5	25.4	24.8	21.3	15.1	12.8	10.2
	5%	WB	8.7	8.0	9.2	10.7	13.6	15.4	16.8	16.8	15.3	12.5	10.7	8.2
		MCDB	9.7	9.3	11.4	14.0	17.9	20.8	23.2	23.0	19.4	14.2	11.8	9.1
	10%	WB	7.9	7.6	8.4	9.9	12.7	14.5	15.9	16.1	14.6	11.8	9.8	7.6
		MCDB	8.8	8.8	10.5	12.6	16.3	18.8	21.2	21.4	18.2	13.5	10.9	8.5
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.9	6.9	8.4	9.6	10.2	11.7	11.7	10.0	7.1	5.5	4.7
		MCDBR	5.1	6.6	8.2	11.0	13.3	14.6	15.9	15.4	13.3	8.7	5.4	4.7
		MCWBR	3.8	4.4	5.1	6.2	7.5	7.6	7.8	7.6	6.7	5.3	3.9	3.6
	5% WB	MCDBR	4.4	5.3	7.3	9.6	11.6	13.0	14.4	13.7	11.2	7.1	4.8	4.3
		MCWBR	3.7	4.0	4.9	5.8	6.9	7.2	7.5	7.2	6.2	4.9	3.8	3.7
Clear-Sky Solar Irradiance	taub	0.318	0.316	0.330	0.355	0.377	0.355	0.348	0.354	0.340	0.331	0.323	0.313	
	taud	2.529	2.558	2.533	2.453	2.407	2.487	2.538	2.521	2.559	2.569	2.524	2.499	
	Ebn at Noon	758	852	892	896	885	906	907	888	870	820	742	711	
	Edh at Noon	61	72	86	103	112	104	98	95	83	71	60	55	
All-Sky Solar Radiation	RadAvg	0.92	1.81	2.83	4.29	5.49	6.00	6.56	5.53	3.93	2.16	1.11	0.74	
	RadStd	0.12	0.24	0.33	0.40	0.43	0.57	0.46	0.36	0.32	0.24	0.17	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.45	N/A	-3.04	+0.98	+0.41	N/A	N/A	-189	+152	+8
(47) Regional (74 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page